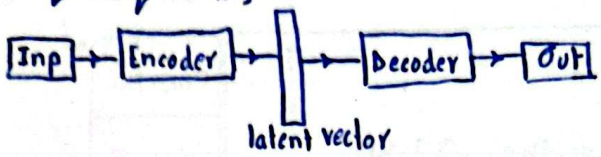
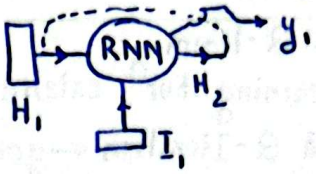
* Seq2Seq models



The encoders & decoders are usually RNNs



* TRASFORMERS FROM SCRATCH

① Input Embeddings:

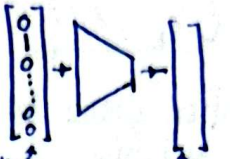
↳ convert a word to a vector

↳ simple MLP : Inp Dim = vocab size

↳ output : d_{model} dim vector

↳ trained on pairwise words to train similar or dissimilar embeddings

↳ embedding formed → needs to be multiplied by $\sqrt{d_{model}}$ → why?



$C = \frac{1}{(b-a)} \Rightarrow \mu = \int p(x)x\,dx \Rightarrow \frac{1}{(b-a)} \frac{x^2}{2}\Big|_a^b = \frac{1}{(b-a)}\left(\frac{b^2-a^2}{2}\right)$

$= \frac{(b+a)}{}$ for $\mu = 0 \Rightarrow b = -a$

$Var = \int p(x)(x-\mu)^2 dx = \int \frac{1}{(b-a)} \left(x - \frac{(b+a)}{2}\right)^2 dx$

$= \frac{1}{(b-a)} \int \left[x^2 + \frac{(b+a)^2}{4} - 2\frac{x(b+a)}{2}\right] dx = \frac{1}{(b-a)}\left[\frac{x^3}{3} + \frac{(b+a)^2}{4}x - \frac{(b+a)x^2}{2}\right]$

$\frac{1}{(b-a)}\left[\frac{(b^3-a^3)}{3} + \frac{(b+a)^2}{4}(b-a) - (b+a)\frac{(b^2-a^2)}{2}\right]$

$= \frac{(a^2+b^2+ab)}{3} + \frac{(b+a)^2}{4} - \frac{(b^2+a^2)}{2} = \frac{9a^2+4b^2+4ab+3b^2+3a^2+6ab}{12}$

$\frac{-(6b^2+6a^2+12ab)}{12} = \frac{a^2+b^2-2ab}{12} = \frac{(a-b)^2}{12} = \sigma^2$

$\sigma^2 = \frac{1}{n} \Rightarrow \frac{(a-b)^2}{12} = \frac{1}{n} \Rightarrow a-b = \frac{2\sqrt{3}}{\sqrt{n}}$

$a = -b$ or $b = -a \Rightarrow 2a = \frac{2\sqrt{3}}{\sqrt{n}} \Rightarrow a = \sqrt{\frac{3}{n}}$

∴ Uniform distribution: $\omega_{ij} \sim U\left[-\sqrt{\frac{3}{fan_{in}}}, \sqrt{\frac{3}{fan_{in}}}\right]$

② Normal distribution $\omega_{ij} \sim N(0, \frac{1}{n})$

How are weights initialized for networks?

↳ weights should not be same: if same then all learn the same features, then pointless

↳ if too small ← vanishing gradient

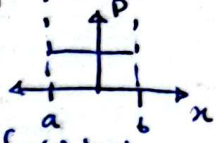
↳ if too large ← exploding gradient

↳ good variance: learns diff features

① Uniform distribution

$\omega_{ij} \sim U\left[\frac{-1}{\sqrt{in}}, \frac{1}{\sqrt{in}}\right]$ why?

a, b



$\int p(x)dx = 1$

$\Rightarrow cx = 1$

$\Rightarrow c(b-a) = 1$

$\Rightarrow c = \frac{1}{(b-a)}$

$\cancel{Var = \int p(x)(x-\mu)^2 dx}$

$\cancel{\mu = \int x p(x) dx = \frac{1}{(b-a)} \frac{x^2}{2}\Big|_a^b}$

$= (b+a)/2$

$\cancel{Var = \frac{1}{(b-a)}\int \left(x - \frac{(b+a)}{2}\right)^2 dx = \frac{1}{(b-a)}\left(\frac{x^3}{3} - \frac{x^2(b+a)}{2} + \frac{(b-a)^3}{4}\right)}$

$\cancel{= \frac{x^3}{3(b-a)} - \frac{x^2}{2} + \frac{(b-a)^2}{4}\Big|_a^b = \frac{(b-a)^2}{3} - \frac{(b-a)^2}{2} + \frac{(b-a)^2}{4}}$

$= \frac{4(b-a)^2 - 6(b-a)^2 + 3(b-a)^2}{12}$

$= \frac{(b-a)^2}{12}$

$Var(y) = \sum_i \omega_i x_i$

$= \sum_i Var(\omega_i x_i)$

$= E(\omega_i^2 x_i^2) - E(\omega_i x_i)^2 = \underbrace{E(\omega_i^2)}_{Var(\omega_i)} \underbrace{E(x_i^2)}_{Var(x_i)} - \underbrace{E(\omega_i)}_{0} \underbrace{E(x_i)}$

$\cancel{= E(\omega_i^2)E(x_i^2)}$

$Var(\omega_i) Var(x_i)$

~~Since all are from the same distribution~~

~~$\therefore Var(y) = n\,Var(\omega)Var(x) \Rightarrow Var(y) \approx Var(x)$~~

~~$\Rightarrow Var(\omega) = \frac{1}{n}$~~

~~to prevent exploding or vanishing~~

~~$\Rightarrow \frac{(b-a)^2}{12} = \frac{1}{n} \Rightarrow b-a = \frac{2\sqrt{3}}{\sqrt{n}}$~~

$\mu = \frac{(b-a)}{2} = 0 \Rightarrow a = b$

③ Xavier / Glorot Normal distribution:

forward pass: $y = \sigma(\omega x + b) \Rightarrow Var(y) = Var(\omega) \times Var(x) \times n$

backward pass $\frac{\partial y}{\partial x} = \frac{\partial(\sigma(z))}{\partial z} \times \omega \Rightarrow Var\left(\frac{\partial y}{\partial x}\right) = Var(\omega) \times n$

$\omega_{ij} \sim U\left[-\sqrt{\frac{6}{n_{in}+n_{out}}}, \sqrt{\frac{6}{n_{in}+n_{out}}}\right]$

$\omega_{ij} \sim N\left[0, \frac{2}{n_{in}+n_{out}}\right]$

∵ $Var(\omega) = \frac{1}{n_{in}}$ & $\frac{1}{n_{out}}$ ↳ harmonic mean taken as compromise : $\sigma^2 = \frac{2}{n_{in}+n_{out}}$

↑ forward pass ↑ backward pass

* Keeps both variance for forward & backward pass in check.

④ He (Kaiming) Initialization: For Relu activation $Var(Relu(x)) \approx \frac{1}{2} Var(x)$

$\Rightarrow \frac{a^2}{3} = \frac{2}{n} \Rightarrow a = \sqrt{\frac{6}{fan_{in}}}$

$\omega_{ij} \sim U\left(-\sqrt{\frac{6}{fan_{in}}}, \sqrt{\frac{6}{fan_{in}}}\right)$ or $\omega_{ij} \sim U\left(0, \frac{2}{fan_{in}}\right)$

CamScanner

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* Weight Initialization:

Technique	Var	Mean	Normal	Uniform
① Le cun general	$\frac{1}{fan_in}$	0	$\sim N\left(0, \frac{1}{fan_in}\right)$	$\sim U\left(-\sqrt{\frac{3}{fan_in}}, \sqrt{\frac{3}{fan_in}}\right)$
② Xavier sigmoid / tanh	$\frac{2}{fan_in + fan_out}$	0	$\sim N\left(0, \frac{1}{n_{in}+n_{out}}\right)$	$\sim U\left(-\sqrt{\frac{6}{n_{in}+n_{out}}}, \sqrt{\frac{6}{n_{in}+n_{out}}}\right)$
③ He ReLU	$\frac{2}{fan_in}$	0	$\sim N\left(0, \frac{2}{n_{in}}\right)$	$\sim U\left(-\sqrt{\frac{6}{n_{in}}}, \sqrt{\frac{6}{n_{in}}}\right)$

② Positional encoding:

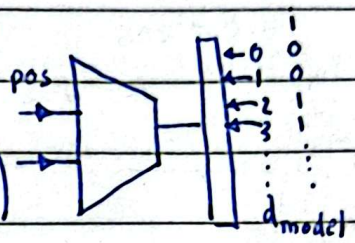
↳ takes the position of a word / element and encodes it as a vector

↳ uses the sin & cosine fnc.

↳ $PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$

↳ $PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$

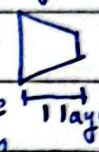
↳ since sin & cosine are used ⇒ var ≈ 0.5



* Why is embedding multiplied by $\sqrt{d_{model}}$?

↳ embedding:

↳ input is one hot encoding



∴ the weights are the output

↳ $Var(y) = Var(w) = \frac{1}{d_{model}}$

↳ why not $\frac{2}{|V| + d_{model}}$ bcuz embedding is a lookup only 1 set of weights active at a time.

∴ to normalize var ≈ 1 (same order of magnitude as pos encoding) multiply by $\sqrt{d_{model}}$ ($Var(kx) = k^2 Var$)

③ Positional encoding + embedding.

$10000^{2i/d_{model}} = e^{2i/d_{model} * \log(10000)}$

④ Normalization: Normalize each vector:

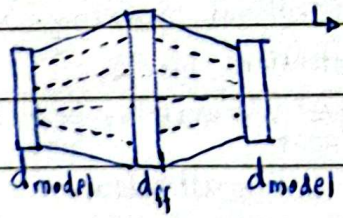
↳ multiplied & added by 2 trainable parameters $x = \alpha\left(\frac{x-\mu}{\sigma}\right) + \beta$

↳ α & β give the network to control the distribution after normalization.

(α ← var, β ← mean)

⑤ Feed Forward: Just a linear MLP with intermediate hidden layer as d_{ff}

↳ ReLU activation applied in the middle



⑥ Attention Block:

↳ Input changed into 3 sets of vectors

↳ (seq, d_{model}) → $\times \boxed{W_K} = \}(seq, d_{model}) = keys$

→ $\times \boxed{W_V} = \}$ = Values

→ $\times \boxed{W_Q} = \}$ = queries

Intuitive Sense:

↳ Query and Keys are compliments and represent how much 2 words correspond to each other

↳ Similarity score given by $QK^T : \sum_{k=1}^{d_m} q_i k_i \Rightarrow Var = d_m \underbrace{Var(q_i)}_{SS} \underbrace{Var(k_i)}_{SS} = d_m Var(inp)^2$

↳ to normalize $\frac{1}{\sqrt{d_{model}}} \times QK^T \Rightarrow Var\left(\frac{QK^T}{\sqrt{d_{model}}}\right) \approx Var(inp)^2 Var(inp)$ $Var(inp)$

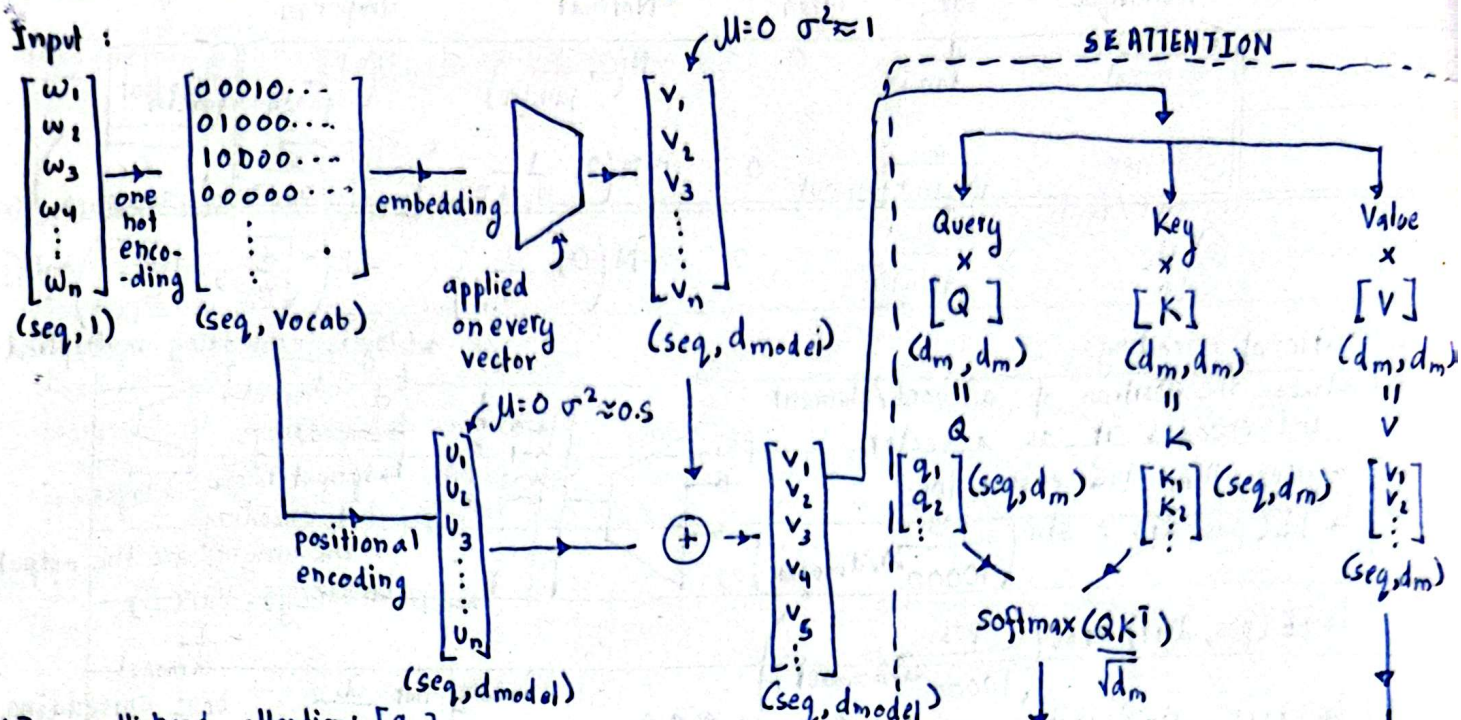
↳ softmax applied to $\frac{QK^T}{\sqrt{d_m}}$ to get values ∈ (0,1) ⇒ weights for the values

↳ $softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) V$

CamScanner

A Batch dimension is added for parallel processing

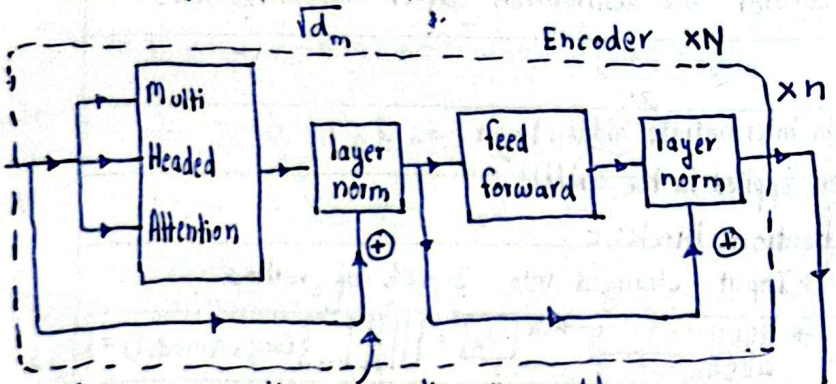
Input :



★In multi head attention: $\begin{bmatrix} q_1 \\ q_2 \\ \vdots \end{bmatrix}$, k, v are sliced along embedding dim with dim d_K ∴ new dimension = (seq, d_K)

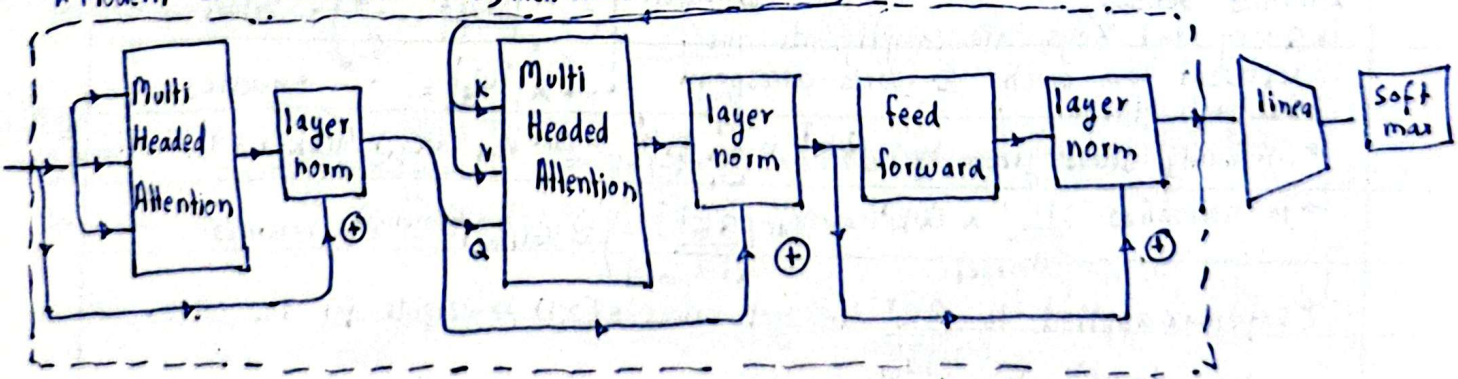
★ a mask can be applied before the softmax
elements to keep = 0 → softmax → same
to not keep = $-\infty$ → 0

∴ scores = $\text{softmax}\left(\frac{QK^T}{\sqrt{d_m}} + M\right)V$



★ Modern convention normalize, then add

★Dropouts:
↳ to prevent overfitting.
ⓐ after positional encoding
ⓑ In attention block
↳ ~~pre~~ post softmax or ~~post~~ pre weighting
ⓒ After entire attention
ⓓ After entire feedforward



Decoder xN

★ L_2 & L_1 Normalization & Regularization:

★ Normalization $\hat{x}_i = \frac{x_i}{\sum |x_i|}$ } L_1 $\hat{x}_i = \frac{x_i}{\sqrt{\sum x_i^2}}$

★Regularization: add penalty according to weights Penalty ∝ $|w_i|$ ← L_1

Loss = Loss + λ Penalty

* L_1: advantage: less severe L_2: more severe but also differentiable : easier to gradient descend.

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* Tokenization:
↳ character level tokenization: small vocab, long sequence
↳ byte pair encoding. (BPE):
↳ ~~creates a binary tree map on top of byte string.~~
↳ algorithm: search for most occurring consecutive pair of chars
↳ replace with new char ← add to vocab
↳ creates a list of merges until target vocabulary size is reached.
↳ Problem: weird inefficient grouping causes inflation in vocab size
↳ solution: use regex to force splitting first & then do BPE

REGEX EXPRESSION IN IN GPT-2:

```
"""'s|'t|'re|'ve|'m|'ll|'d|...?\p{L}+|...?\p{N}+|...?[^\s\p{L}\p{N}]+|\s+(?!\S)|\s+"""
```

- `'s|'t|'re|'ve|'m|'ll|'d` — postfix expr (↑ make sure iss not case sensitive)
- `...?` — optional space
- `\p{L}+` — letters
- `...?` — optional space
- `\p{N}+` — numbers
- `...?` — optional space
- `[^\s\p{L}\p{N}]+` — not letters/numbers/whitespace
- `\s+(?!\S)` — spaces except the last
- `\s+` — trailing spaces

* usually max 1-3 digits of numbers in tokenization in GPT-4

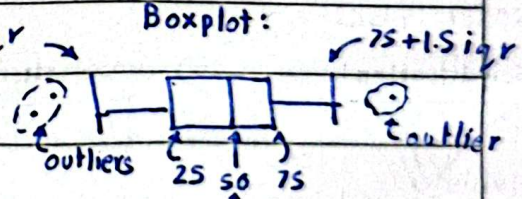
tokens
↕
BPE tokenizer
↕
Byte encoding
↕
string

* Special tokens:
↳ <|end of text|> token ← marks the end of training text
↳ instructs model to wipe memory ← intuitively
↳ tiktoken ← openAI tokenizer
↳ can be extended to have multiple special tokens to have special meanings.

* Regex Tokenizer: First splits according to regex & then applies function on each segment.

* Fill missing data:
* Fill Outliers:
↳ statistical: calculate percentiles
↳ calculate iqr: interquartile range
↳ 75 pct − 25 pct = iqr
↳ outliers ∈ $(0, 25\text{pct} - 1.5\text{iqr}) \cup (75\text{pct} + 1.5\text{iqr}, \infty)$

Boxplot:



residuals.
↳ diff btw predicted & actual
↳ standardized: $\frac{\text{residual}}{\sigma \text{ of residuals}}$
↳ deleted: delete current obs, recalculate residuals
↳ Studentized: deleted residuals / standard dev

} stuff to do: delete, transform, replace, predict

* COVARIANCE: relation between 2 variables: $E((x - E(x))(y - E(y))) = E(xy - xE(y) - yE(x) + E(x)E(y))$
$= E(XY) - E(X)E(y) - E(X)E(y) + E(x)E(y) = E(XY) - E(X)E(Y)$

* Pearson's correlation coefficient: $\frac{\cancel{Var}\ Cov(X,Y)}{\sigma_x \sigma_y} \in [-1, 1]$

* Skewness: to judge assymmetry: $\frac{\frac{1}{n}\sum(x_i - \bar{x})^3}{\left(\frac{1}{n}\sum(x_i - \bar{x})^2\right)^{3/2}}$ ← σ^3